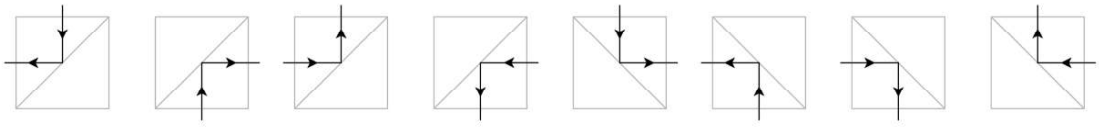
Problem 07: Robot Maze

Time limit: 5 seconds

A robot must travel through a rectangular grid. Some cells contain mirrors. These mirrors reflect deadly laser beams fired from fixed laser sources. A laser beam travels **in all 4 cardinal directions from its starting point** in a straight line until it hits a wall or leaves the grid. Any cell touched by a beam becomes instantly unsafe for the robot. The robot can move between adjacent cells (sharing a side) that are safe. Stepping on any cell hit by a laser results in immediate termination

Each mirror has 2 possible configurations: / or \ (Explained in the figure below).

To avoid the deadly laser, the robot can rotate a mirror with a cost of **1**. It can rotate multiple mirrors or none.



The above shows the effect of a laser when it passes through mirrors in / (first 4 figures) and \ (last 4 figures).

A mirror configuration is **valid** if there exists at least one fully safe path from the robot's start cell to its target cell.

Your task is to compute the following:

- The number of valid mirror configurations
- The minimum cost of any valid mirror configuration – If none, output **-1**

Description of objects in the grid:

- . — empty cell
- # —brick wall
- S — robot start position
- T — robot target position
- L — laser source
- / — mirror (forward slash)
- \ — mirror (backslash)

Input:

The first line contains 2 space separated integers N (5 <= N <= 50) and M (5 <= N <= 50) – the number of rows and columns in the grid respectively.

Then N lines follow. Each line has M characters, where each character denotes an object in the grid.

It is guaranteed that there is a unique S, T, L in the entire N*M total characters.

It is also guaranteed that there are no more than 15 mirrors.

Output:

Print 2 integers, on the same line. The first integer is the number of valid mirror configurations. The second integer is the minimum cost of any valid mirror configuration.

Sample Input 1	Sample Output 1
5 7 ..... .S.#.L. ..... .\.../. ....T..	2 1

Sample Input 2	Sample Output 2
6 6 ..... .S../. ..#L#. ..##.. ./..T.	4 0